

```
In [50]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

trades = pd.read_csv("D:\historical_data.csv")
sentiment = pd.read_excel("D:\sentiment_index.xlsx")

In [71]: print(trades["timestamp"].head(10))

0    02-12-2024 22:150
1    02-12-2024 22:150
2    02-12-2024 22:150
3    02-12-2024 22:150
4    02-12-2024 22:150
5    02-12-2024 22:150
6    02-12-2024 22:150
7    02-12-2024 22:150
8    02-12-2024 22:150
9    02-12-2024 22:150
Name: timestamp, dtype: object

In [101]: trades["timestamp"] = pd.to_datetime(trades["timestamp"])
trades["date"] = trades["timestamp"].dt.date
sentiment["date"] = pd.to_datetime(sentiment["date"])
sentiment["date"] = sentiment["date"].dt.date

In [111]: merged = pd.merge(
    trades,
    sentiment[["date", "classification", "value"]],
    on="date",
    how="left",
)

In [109]: merged.head()

In [102]:
```

	Account	Coin	Execution Price	Size USD	Size Tokens	Side	Timestamp IST	Start Position	Direction	Closed PnL	Transaction Hash	Order ID	Crossed	Fee	Trade ID	Timestamp	Date	classification	value
0	0x6a5eacfa92b011f6530346d2c192a0464082ed	@107	7.9769	986.87	7872.16	BUY	2024-02-12 22:50:00	0.000000	Buy	0.0	0x609451986a1874ca3a9804184126a2021f500c95bac...	5201770630	True	0.345404	8.960000e+14	1.730000e+12	2024-02-12	Greed	70.0
1	0x6a5eacfa92b011f6530346d2c192a0464082ed	@107	7.9800	16.00	127.68	BUY	2024-02-12 22:50:00	986.524596	Buy	0.0	0x609451986a1874ca3a9804184126a2021f500c95bac...	5201770630	True	0.050601	4.430000e+14	1.730000e+12	2024-02-12	Greed	70.0
2	0x6a5eacfa92b011f6530346d2c192a0464082ed	@107	7.9855	144.09	1150.63	BUY	2024-02-12 22:50:00	1002.518996	Buy	0.0	0x609451986a1874ca3a9804184126a2021f500c95bac...	5201770630	True	0.050431	6.800000e+14	1.730000e+12	2024-02-12	Greed	70.0
3	0x6a5eacfa92b011f6530346d2c192a0464082ed	@107	7.9874	184.98	1142.04	BUY	2024-02-12 22:50:00	1146.508554	Buy	0.0	0x609451986a1874ca3a9804184126a2021f500c95bac...	5201770630	True	0.050403	1.080000e+15	1.730000e+12	2024-02-12	Greed	70.0
4	0x6a5eacfa92b011f6530346d2c192a0464082ed	@107	7.9894	8.73	69.75	BUY	2024-02-12 22:50:00	1289.488521	Buy	0.0	0x609451986a1874ca3a9804184126a2021f500c95bac...	5201770630	True	0.003055	1.050000e+15	1.730000e+12	2024-02-12	Greed	70.0

```
In [131]: print(merged.shape)
print(merged.info())

(66359, 19)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 66359 entries, 0 to 66358
Data columns (total 19 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   Account     66359 non-null    object
 1   Coin        66359 non-null    object
 2   Execution Price  66359 non-null    float64
 3   Size Tokens  66359 non-null    float64
 4   Size USD     66359 non-null    float64
 5   Side         66359 non-null    object
 6   Timestamp IST  66359 non-null    datetime64[ns]
 7   Start Position  66359 non-null    float64
 8   Direction     66359 non-null    object
 9   Closed PnL    66359 non-null    float64
10  Transaction Hash  66359 non-null    object
11  Order ID     66359 non-null    int64
12  Crossed      66359 non-null    bool
13  Fee          66359 non-null    float64
14  Trade ID    66359 non-null    float64
15  Timestamp    66359 non-null    float64
16  Date         66359 non-null    object
17  classification  57258 non-null    object
18  value        57258 non-null    float64
dtypes: bool(1), datetime64[ns](1), float64(9), int64(1), object(7)
memory usage: 9.2+ MB
None

In [141]: merged.issmall().sum()

Out[141]:
Account      0
Coin          0
Execution Price  0
Size Tokens   0
Size USD      0
Side          0
Timestamp IST  0
Start Position  0
Direction     0
Closed PnL    0
Transaction Hash  0
Order ID      0
Crossed       0
Fee           0
Trade ID      0
Timestamp     0
Date          0
classification 9101
value         9101
Dtype: int64

In [151]: plt.figure(figsize=(6,4))
sns.countplot(data=merged, x="classification")
plt.xticks(rotation=45)
plt.show()
```

```
In [161]: merged.groupby("classification")["Closed PnL"].mean().sort_values()

Out[161]:
classification
Neutral      58.574302
Greed       61.412296
Fear       135.148480
Extreme Greed 177.042714
Extreme Fear 346.042589
Name: Closed PnL, dtype: float64

In [171]: plt.figure(figsize=(7,5))
sns.boxplot(
    x="classification",
    y="Closed PnL",
    data=merged
)
plt.xticks(rotation=45)
plt.show()
```

```
In [181]: merged.groupby("classification")["Closed PnL"].sum()

Out[181]:
classification
Extreme Fear  1.570806e+05
Extreme Greed  1.502091e+06
Fear         2.124941e+06
Greed        1.202224e+06
Neutral      7.385942e+05
Name: Closed PnL, dtype: float64

In [191]: merged["Min"] = merged["Closed PnL"] > 0
merged.groupby("classification")["Min"].mean()*100

Out[191]:
classification
Extreme Fear  32.444444
Extreme Greed  49.039636
Fear         44.401666
Greed        43.113222
Neutral      65.708204
Name: Min, dtype: float64

In [201]: pd.crosstab(
    merged["Side"],
    merged["classification"],
)

Out[201]:
classification  Extreme Fear  Extreme Greed  Fear  Greed  Neutral
Side
BUY              340          3741  7497   9482   6419
SELL             110          5165  8300  10099   6105

In [211]: sns.countplot(
    data=merged,
    x="Side",
    hue="classification",
)

Out[211]: <axes: xlabel='Side', ylabel='count'>
```

```
In [231]: merged.groupby("classification")["Size USD"].sum()

Out[231]:
classification
Extreme Fear  2.990863e+06
Extreme Greed  2.027434e+07
Fear         2.124941e+06
Greed        1.202224e+06
Neutral      7.385942e+05
Name: Size USD, dtype: float64

In [241]: merged.groupby("classification")["Size USD"].mean()

Out[241]:
classification
Extreme Fear  666.116933
Extreme Greed  7890.436629
Fear         14847.015399
Greed        12121.704719
Neutral      8037.955989
Name: Size USD, dtype: float64

In [251]: merged.groupby("classification")["Fee"].mean()

Out[251]:
classification
Extreme Fear  1.292958
Extreme Greed  2.022846
Fear         2.360883
Greed        2.444041
Neutral      1.476051
Name: Fee, dtype: float64

In [261]: merged["profit"] = merged.groupby(
    ["Coin", "classification"],
    as_index=False,
)["Closed PnL"].sum()

Out[261]:
Coin  classification
P107  Greed          0.000000e+00
P107  Extreme Fear  8.000000e+00
P107  Extreme Greed  1.433601e+06
P107  Fear          1.365117e+04
P107  Greed         1.394817e+05
P107  Neutral       7.240208e+03
P107  Extreme Greed  6.255476e+03
P107  Extreme Greed  4.724020e+03
P107  Extreme Greed  4.013288e+02
P107  Greed         0.000000e+00
Name: Closed PnL, Length: 381, dtype: float64

In [271]: merged["profit"].sort_values(ascending=False).head(10)

Out[271]:
Coin  classification
P107  Extreme Greed  1.433601e+06
P107  Greed         1.394817e+05
BTC  Fear          4.181128e+05
BTC  Fear          2.944589e+05
BTC  Greed         2.037024e+05
BTC  Greed         1.904219e+05
BTC  Greed         1.717619e+05
BTC  Greed         1.630139e+05
BTC  Greed         1.455436e+05
BTC  Neutral       1.392438e+05
BTC  Extreme Fear  1.255130e+05
BTC  Greed         1.084271e+05
BTC  Extreme Greed  7.257379e+04
BTC  Greed         6.821170e+04
BTC  Greed         6.311704e+04
BTC  Neutral       5.812157e+04
BTC  Extreme Greed  4.604930e+04
Name: Closed PnL, dtype: float64

In [281]: top = merged.groupby("Account")["Closed PnL"].sum()
top.sort_values(ascending=False).head(10)

Out[281]:
Account
0x81211a4a2022276a30a7a2f183a0b0e8d3  2.143338e+06
0x313818979e0cf1999169a3b3e0c345f2a9012  1.600230e+06
0x13a032f6a77b0a512a4e24a470240e4f11  8.404226e+05
0x31039141d0b0a2c277650330a5818b0e7113  4.145413e+05
0x72c48424e1dffc72e46d00d0e0a6498cf92a0  4.030113e+05
0x707770e5a030b0e78705313a0a5a1170a68  3.780546e+05
0x4f91e4d9b70a1024f981a404a5b27b9e9760  3.089759e+05
0x403a0a0eb08a0305a4e0a0b0a040402404041  1.995546e+05
0x4d84a931d2028f5d8e4092a76885318264a  1.087312e+05
0x4d84a931d2028f5d8e4092a76885318264a  6.745524e+04
Name: Closed PnL, dtype: float64

In [291]: top.sort_values(ascending=False).head(10).plot(
    kind="bar",
    figsize=(10,5),
)

Out[291]: <axes: xlabel='Account',>
```

```
In [317]: numeric = merged.select_dtypes(include=np.number)
plt.figure(figsize=(10,8))
sns.histplot(
    numeric.sort(),
    x="value",
    cmap="coolwarm",
)

Out[317]: <axes: >
```

```
In [327]: plt.figure(figsize=(8,5))
sns.histplot(
    merged["Closed PnL"],
    bins=50
)

Out[327]: <axes: xlabel='Closed PnL', ylabel='Count'>
```

```
In [337]: merged["hour"] = merged["timestamp"].dt.hour
merged.groupby("hour")["Closed PnL"].mean().plot(figsize=(10,5))

Out[337]: <axes: xlabel='hour',>
```

```
In [347]: merged.groupby("classification").size()

Out[347]:
classification
Extreme Fear    450
Extreme Greed   8906
Fear           15797
Greed          19181
Neutral        12524
Dtype: int64

In [357]: merged.groupby("Side")["Closed PnL"].mean()

Out[357]:
Side
buy    97.340616
sell   102.97129
Name: Closed PnL, dtype: float64

In [367]: merged.nlargest(10, "Closed PnL")

Out[367]:
```

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL	Order ID	Crossed	Fee	Trade ID	Timestamp	Date	classification	value	Win	Hour	
18045	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	1570.10000	186.5206	292870.12	BUY	2025-12-04 12:31:00	-303.3564	Close Short	193329.09010	...	85971520603	False	29.287012	1.890000e+15	1.740000e+12	2025-12-04	NaN	NaN	True	12
12720	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2294.00000	300.0000	685200.00	BUY	2025-02-28 01:52:00	-1300.0000	Close Short	115287.00010	...	75527957029	False	68.520000	8.890000e+14	1.740000e+12	2025-02-28	Extreme Fear	16.0	True	1
18043	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	1570.10000	108.4516	170279.86	BUY	2025-04-02 12:22:00	-311.8148	Close Short	78862.72032	...	85971520603	False	17.027985	6.760000e+14	1.740000e+12	2025-04-02	NaN	NaN	True	12
18024	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	1570.10000	99.7612	156630.06	BUY	2025-04-02 11:58:00	-417.5139	Close Short	72377.74821	...	85971520603	False	15.663008	3.860000e+14	1.740000e+12	2025-04-02	NaN	NaN	True	12
16328	0x083384897e0f19899168a3b3e0c345f2a9012	SOL	210.88000	14331.4100	3071855.74	BUY	2025-04-02 10:36:00	-5999.0800	Close Short	71535.71674	...	69292154528	True	90.596722	9.360000e+14	1.740000e+12	2025-04-02	Fear	44.0	True	10
16377	0x083384897e0f19899168a3b3e0c345f2a9012	SOL	202.89000	866.1300	175555.89	BUY	2025-04-02 11:15:00	-1689.3800	Close Short	50379.05774	...	69304541930	False	8.777794	4.660000e+14	1.740000e+12	2025-04-02	Fear	44.0	True	11
49084	0x12314e40d022276a30a7a2f183a0b0e8d3	DOGE	0.25415	236789.0000	60179.82	BUY	2025-04-02 21:56:00	-286901.0000	Close Short	48504.90555	...	70949191826	True	21.062973	9.990000e+14	1.740000e+12	2025-04-02	NaN	NaN	True	21
17075	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2541.70000	308.1061	786664.97	BUY	2025-02-25 01:37:00	-1504.5823	Close Short	47883.82595	...	74570997088	True	274.979241	9.100000e+13	1.740000e+12	2025-02-25	Fear	25.0	True	3
45581	0x12314e40d022276a30a7a2f183a0b0e8d3	@107	17.81000	2483.0700	44223.48	SELL	2024-12-12 15:58:00	67406.2600	Sel	44223.45187	...	54838005127	True	15.478216	1.020000e+15	1.730000e+12	2024-12-12	Extreme Greed	83.0	True	15
15353	0x083384897e0f19899168a3b3e0c345f2a9012	SOL	202.89000	736.7700	149355.91	BUY	2025-04-02 11:18:00	-2876.6800	Close Short	42854.74279	...	69304541930	False	7.469736	8.390000e+14	1.740000e+12	2025-04-02	Fear	44.0	True	11

```
In [367]: merged.nsmall(10, "Closed PnL")

Out[367]:
```

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp IST	Start Position	Direction	Closed PnL	Order ID	Crossed	Fee	Trade ID	Timestamp	Date	classification	value	Win	Hour	
14682	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	4041.0000	201.5650	814524.31	BUY	2024-06-12 23:15:00	-559.28340	Close Short	-117990.10410	...	53252410400	False	81.452416	1.060000e+15	1.730000e+12	2024-06-12	Greed	72.0	False	23
3737	0x513b82c9a6778a581a244a30a45b278e9760	BTC	48243.0000	4.5880	373314.16	BUY	2025-08-04 00:37:00	-7.79714	Close Long	-29370.11890	...	85369144282	False	35.838158	1.000000e+14	1.740000e+12	2025-08-04	NaN	NaN	False	23
25404	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2970.0000	866.1300	178200.00	SELL	2024-04-14 01:37:00	88.82000	Close Long	-21524.46000	...	7983732002	False	32.370000	1.060000e+14	1.740000e+12	2024-04-14	Greed	72.0	False	1
3566	0x513b82c9a6778a581a244a30a45b278e9760	HYPER	10.4970	9261.7300	97220.38	BUY	2025-07-04 00:37:00	22874.42000	Close Long	-21133.46888	...	84309081969	False	9.333155	5.170000e+14	1.740000e+12	2025-07-04	NaN	NaN	False	8
15760	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2535.0000	166.1300	398960.07	SELL	2025-03-02 08:48:00	1232.26000	Close Long	-19841.26314	...	68719119361	True	19.847903	4.700000e+14	1.740000e+12	2025-03-02	Fear	26.0	False	0
25299	0x083384897e0f19899168a3b3e0c345f2a9012	BTC	91627.0000	2.8233	256690.51	SELL	2025-10-01 00:37:00	3.76307	Close Long	-18993.46842	...	61875044905	True	90.541678	8.880000e+14	1.740000e+12	2025-10-01	NaN	NaN	False	2
14696	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	4041.0000	32.2018	130127.47	BUY	2024-06-12 23:15:00	-313.20180	Close Short	-18849.96767	...	53252410400	False	13.012747	6.290000e+14	1.730000e+12	2024-06-12	Greed	72.0	False	23
23068	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2657.6000	42.9605	111779.38	SELL	2024-04-08 23:14:00	93.28910	Close Long	-18360.67007	...	32439862244	True	39.122994	5.170000e+14	1.720000e+12	2024-04-08	Extreme Greed	76.0	False	23
23952	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2660.0000	41.8621	111378.30	SELL	2024-04-08 23:14:00	185.21180	Close Long	-18148.47621	...	32439862244	True	38.982408	3.360000e+14	1.720000e+12	2024-04-08	Extreme Greed	76.0	False	23
23089	0x083384897e0f19899168a3b3e0c345f2a9012	ETH	2654.5000	40.9690	103753.01	SELL	2024-04-08 23:14:00	40.96900	Close Long	-18011.33336	...	32439862244	True	38.035552	5.240000e+14	1.720000e+12	2024-04-08	Extreme Greed	76.0	False	23

```
In [367]: 10 rows * 21 columns
```